Monodromy Theory for the Hénon Map

$$H_{a,c} : \begin{pmatrix} x \\ y \end{pmatrix} \longmapsto \begin{pmatrix} x^2 + c - ay \\ x \end{pmatrix}$$

parameter space $\mathbb{C}^2 \ni (a, c)$

phase space $\mathbb{C}^2 \ni (x, y)$

$$K_{a,c} := \left\{ (x,y) \in \mathbb{C}^2 \;\middle|\; \left\{ H_{a,c}^{\circ n}(x,y) \right\}_{n \in \mathbb{Z}} \text{ is bdd} \right\}$$

$$\mathcal{H} := \left\{ (a,c) \in \mathbb{C}^2 \;\middle|\; K_{a,c} \text{ is conj to } \Sigma_2^{\mathbb{Z}} \right\}$$

(& hyperbolic)

$$\mathcal{H} \cap \{ a = 0 \} = \mathbb{C} \setminus M$$

$$\rho : \pi_1(\mathcal{H}, *) \longrightarrow \mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$$

$$\Gamma := \mathrm{im}\, \rho \subset \mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$$

Hubbard's Conjecture

σ: shift map

$\Gamma \cup \{\sigma\}$ generate $\mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$

☆ $\Gamma \neq \{ id \}$, $\exists \phi \in \Gamma$: infinite order

already shown

Today $\Gamma \subset \mathrm{Inert}(\Sigma_2)$ provided

$\exists \infty$ly many non-Wieferich prime numbers.

FACT $\mathrm{Aut}(\Sigma_2^{\mathbb{Z}}) = \mathbb{Z}\langle\sigma\rangle \oplus \mathrm{Inert}(\Sigma_2)$

$\phi \in \mathrm{Inert}(\Sigma) \Leftrightarrow \phi$ satisfies SGCC

SGCC

☆ $S_n(\phi) \in \mathbb{Z}_2$ sign number

$S_n(\phi) = 1$ iff the sign of the permutation induced by ϕ on the set of periodic bits of least period n.

☆ $g_n(\phi) \in \mathbb{Z}_n$ gyration number

For each per. orbit $\mathcal{U}$ of least period n, choose an arbitrary base pt $x_{\mathcal{U}} \in \mathcal{U}$

Since $\phi(x_{\mathcal{U}})$ & $x_{\phi(\mathcal{U})}$ are the same orbit,

$\exists\, k(\mathcal{U})$ s.t. $\phi(x_{\mathcal{U}}) = \sigma^{k(\mathcal{U})}(x_{\phi(\mathcal{U})})$

$g_n(\phi) = \sum_{\mathcal{U}} k(\mathcal{U}) \mod n$

☆ ϕ satisfies $SGCC_n$

$\iff \quad g_n(\phi) + \sum_{i>0} S_{\frac{n}{2^i}}(\phi) = 0$

- $S_{\frac{n}{2^i}}(\phi) = 0$ when $\frac{n}{2^i} \notin \mathbb{N}$
- $S_{\frac{n}{2^i}}(\phi) \in \mathbb{Z}_2 \cong \{0, \frac{n}{2}\} \subset \mathbb{Z}_n$

☆ ϕ satisfies SGCC

$\iff$ ϕ satisfies SGCC for all $n \geq 2$

☆ σ does not. $S_1(\sigma) = 0$, $g_2(\sigma) = 1$

☆ compositions of finite order auto satisfy SGCC

☆ p: odd prime

$SGCC_p(\sigma) = 0 \iff p$ is Wieferich

$S_{\frac{p}{2^i}}(\sigma) = 0$ since p is odd.

# of orbits $= (2^p - 2)/p$.

each orbit is gyrated by one,

$\Rightarrow g_p(\sigma) = (2^p-2)/p \bmod p \equiv 0$ iff

p is Wieferich

$M_1 = \{(a,c) \mid H_{a,c}$ has $\exists$ att. fixed pt $\}$

$\exists$ repelling

$M_1 \cap \{a=0\}$ = main cardioid. $M_1 \cap \mathcal{H} = \emptyset$

$\begin{pmatrix} x \\ y \end{pmatrix}$ is fixed $\iff$ $x = x^2 + c - ay$, $y = x$

$\iff$ $x^2 - (1+a)x + c = 0$, $x = y$

eigen value $\lambda_\pm = x \pm \sqrt{x^2 - a}$

$\lambda_+ \cdot \lambda_- = a$

$\begin{pmatrix} 2x & -a \\ 1 & 0 \end{pmatrix}$

$\Downarrow$

$\begin{vmatrix} \lambda - 2x & a \\ -1 & \lambda \end{vmatrix}$

$\shortparallel$

$\lambda^2 - 2x\lambda + a = 0$

$\Downarrow$

$x = \frac{1}{2}\left(\lambda + \frac{a}{\lambda}\right)$

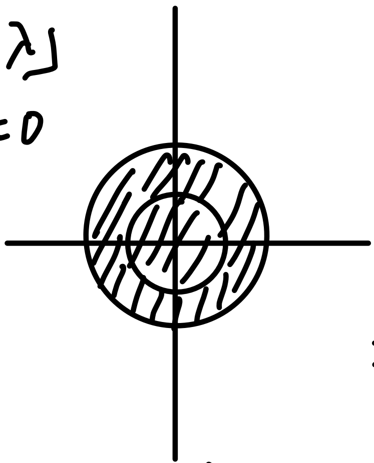
Fix a: eigs correspond to M_1

$1 \le |\lambda| \le |a|$ or

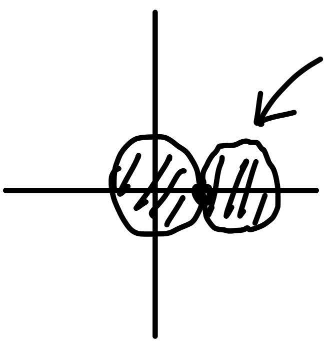
$|a| \le |\lambda| \le 1$

λ]

$a=0$



x]



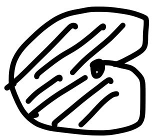
the other fixed pt is attracting.

c]

$x_p + x_q = 1 + a$

$a \neq 0$

$x_q = 1 + a - x_p$



$c = x^2 + (1+a)x$

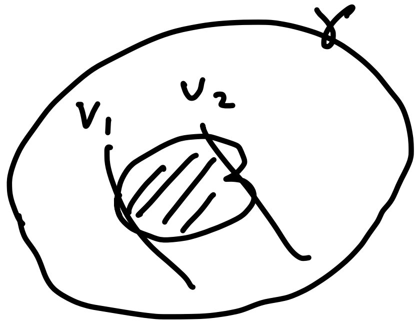
2:1 covering except at s.v.

Lemma $\phi \in \Gamma \Rightarrow SGCC_2(\phi) = 0$

$\because$ $\rho(\gamma) = \phi$, $\gamma \in \pi_1(\mathcal{H})$

$V_1 = \{ c = \frac{1}{4}(a+1)^2 \}$

$V_2 = \{ c = -\frac{3}{4}(a+1)^2 \}$



V_1 and V_2 are homotopic in $M \subset \mathbb{C}^2 \setminus H$

action on the periodic pt

every time γ winds around V_1,
$\rho(\gamma)$ switches two fixed pts

everytime γ winds around V_2,
$\rho(\gamma)$ gyrate the periodic pt.

$S_1(\phi) = g_2(\phi) = 1$ iff # winding is odd

$S_1(\phi) = g_2(\phi) = 0$ iff # even

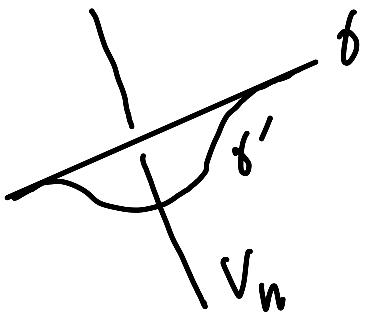
<u>Lemma</u> $\phi C \in \Gamma \Rightarrow SGCC_p(\phi) = 0$ for prime p.

$\rho(\gamma) = \phi$ Need to show $g_p(\phi) = 0$.

<u>Step 1</u> if $\gamma \sim \gamma'$ in $\mathbb{C}^2 \setminus M_1$

then $g_p(\rho(\gamma)) = g_p(\rho(\gamma'))$

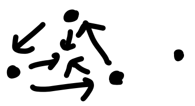
∵



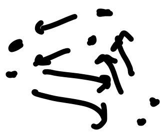
We need to prove that no gyration happens around $V_n \cap (\mathbb{C}^2 \setminus M_1)$

even on V_n, Solutions of $\{ \text{—} \}$ can be either of $\underbrace{\text{period}}_{\text{least}}$ 1 or p.

period multiplying

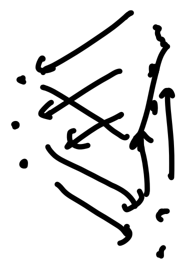


or



saddle-node type

In $\mathbb{C}^2 \setminus M_1$, only period p case can happen since all the fixed pts are hyp.



We only need to care about collisions of two orbits.

$\{\text{triple collision locus}\} =: V_p' \subset V_p$

$\Uparrow$

sub affine algebraic variety of dim 0.

Since $V_p' \cap \{a = 0\} = \emptyset$

Pairwise interchange of orbits does not contribute to gyration number.

$$\Rightarrow g_p(\rho(\gamma')) = g_p(\rho(\gamma))$$

<u>Step 2</u> γ_0 = loop around the Mandelbrot set

$\exists k, \quad \gamma^2 \underset{\text{conj}}{\sim} \gamma_0^{2k}$ in $\mathbb{C}^2 \setminus M_1$

$\because$ γ^2 defines a loop in x-space which is homotopic to $\mathbb{C} \setminus M$

<u>Step 3</u> $g_p(\rho(\gamma_0)) = 0$

$$\Rightarrow g_p(\rho(\gamma_0^{2k})) = 0$$

$$\Rightarrow g_p(\rho(\gamma^2)) = 2\, g_p(\rho(\gamma)) = 0$$

$$\Rightarrow g_p(\rho(\gamma)) = 0$$

<u>THM</u> $\phi \in \Gamma \Rightarrow \phi$ satisfies SGCC.

Decompose ϕ as $\phi = \sigma^k \phi'$

where $\phi' \in \mathrm{Inert}(\Sigma_2^{\mathbb{Z}})$

Assume $k \neq 0$

Choose a non-Wieferich prime p

which is coprime to k

Then $SGCC_p(\sigma^k) = k\, SGCC_p(\sigma)$

is nonzero since $SGCC_p(\sigma) \neq 0$

and k is coprime to p.

$$\underset{\substack{\| \\ 0 \\ \text{by Lemma}}}{SGCC_p(\phi)} = \underset{\substack{\neq \\ 0}}{SGCC_p(\sigma^k)} + \underset{\substack{\| \\ 0 \\ \phi' \text{ is inert}}}{SGCC(\phi')}$$

$\Rightarrow$ Contradiction